Light

GCSE Computer Science — Cyber Security Threats (Set 3)

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

Malware spreads to a user's computer when they visit a compromised website, without clicking or downloading anything.
What is this infection method called?

- ○ **A.** A drive-by download
- ○ **B.** A brute force attack
- ○ **C.** Social engineering
- ○ **D.** Packet sniffing

Q2 (1 mark)

A security flaw in software is discovered and exploited by attackers before the software developer knows about it.
What is this type of vulnerability called?

- ○ **A.** A legacy system
- ○ **B.** A zero-day vulnerability
- ○ **C.** An open-source project
- ○ **D.** A beta version

Q3 (1 mark)

An employee deliberately copies confidential company files to a personal USB drive before leaving for a competitor.
What type of security threat is this?

- ○ **A.** A phishing attack
- ○ **B.** A DDoS attack
- ○ **C.** An insider threat
- ○ **D.** A man-in-the-middle attack

Q4 (1 mark)

A website asks users to identify traffic lights in a grid of images before creating an account.
What is the purpose of this security measure?

- ○ **A.** To encrypt the user's password
- ○ **B.** To compress images for faster loading
- ○ **C.** To test the user's eyesight
- ○ **D.** To verify the user is human and not an automated bot

Q5 (1 mark)

Attackers use stolen username and password combinations from one website to try logging into many other websites.
What is this attack called?

- ○ **A.** Credential stuffing
- ○ **B.** Packet sniffing
- ○ **C.** SQL injection
- ○ **D.** Defragmentation

Q6 (1 mark)

A padlock icon appears in the browser's address bar when visiting a banking website.
What does this indicate?

- ○ **A.** The website contains no advertisements
- ○ **B.** The website has no viruses
- ○ **C.** The user's computer has antivirus software
- ○ **D.** The connection is encrypted using HTTPS

Q7 (1 mark)

A company separates its guest WiFi network from its internal business network.
What security benefit does this provide?

- ○ **A.** It makes the guest WiFi faster

- ○ **B.** It allows guests to print to company printers
- ○ **C.** It reduces the electricity costs
- ○ **D.** It prevents visitors from accessing internal systems if guest devices are compromised

Q8 (1 mark)

Attackers compromise a website frequently visited by employees of a specific company, hoping to infect their computers.
What type of attack is this?

- ○ **A.** A brute force attack
- ○ **B.** A watering hole attack
- ○ **C.** A denial of service attack
- ○ **D.** Shoulder surfing

Q9 (1 mark)

An external company reviews an organisation's security policies, procedures, and technical controls.
What is this process called?

- ○ **A.** A security audit
- ○ **B.** A phishing test
- ○ **C.** Data compression
- ○ **D.** System defragmentation

Q10 (1 mark)

A disgruntled employee writes code that will delete all company files on a specific future date.
What type of malware is this?

- ○ **A.** A worm
- ○ **B.** Adware
- ○ **C.** A logic bomb
- ○ **D.** Spyware

Q11 (1 mark)

A system administrator only grants users access to the specific files and systems they need for their job.
What security principle is being applied?

- ○ **A.** Defence in depth
- ○ **B.** Security through obscurity
- ○ **C.** Open access policy
- ○ **D.** The principle of least privilege

Q12 (1 mark)

An attacker registers the domain 'amazOn.com' (with a zero) hoping users will mistype 'amazon.com'.
What is this technique called?

- ○ **A.** Packet sniffing

- B. SQL injection
- C. Typosquatting
- D. Defragmentation

Q13 (2 marks)

A type of malware changes its code each time it replicates, making it difficult for antivirus software to detect using signatures.
What is this type of malware called?

- A. Adware
- B. Polymorphic malware
- C. Open-source software
- D. Firmware

Q14 (2 marks)

An attacker intercepts communication between a user and their bank, reading and modifying data in both directions without either party knowing.
What type of attack is this?

- A. A denial of service attack
- B. A brute force attack
- C. Shoulder surfing
- D. A man-in-the-middle attack

Q15 (2 marks)

A network security system monitors traffic for suspicious patterns and automatically blocks potentially malicious connections.
What type of system is this?

- ○ **A.** An Intrusion Prevention System (IPS)
- ○ **B.** A word processor
- ○ **C.** A spreadsheet application
- ○ **D.** A defragmentation tool

Q16 (2 marks)

A company sets up a fake server containing apparent valuable data to attract and study attackers.
What is this security technique called?

- ○ **A.** Load balancing
- ○ **B.** Data compression
- ○ **C.** A honeypot
- ○ **D.** Defragmentation

Q17 (2 marks)

A software development team reviews code for security vulnerabilities before releasing new features.
Why is this practice important?

- ○ **A.** It makes the software run faster

- ○ **B.** It reduces the file size of the application
- ○ **C.** It identifies and fixes security flaws before attackers can exploit them in production
- ○ **D.** It removes the need for user passwords

Q18 (2 marks)

A company creates a detailed plan for restoring IT systems after a major cyber attack or natural disaster.
What is this plan called?

- ○ **A.** An acceptable use policy
- ○ **B.** A privacy notice
- ○ **C.** A software licence
- ○ **D.** A disaster recovery plan

Q19 (2 marks)

An attacker steals a user's session cookie after they log into a website, allowing the attacker to impersonate them without knowing their password.
What is this attack called?

- ○ **A.** Session hijacking
- ○ **B.** Defragmentation
- ○ **C.** Data compression
- ○ **D.** Load balancing

Q20 (2 marks)

A company requires all employees to complete annual cyber security training covering topics like phishing recognition and password hygiene. What is the primary purpose of this training?

- ○ **A.** To teach employees how to write computer programs
- ○ **B.** To reduce human error and improve employee ability to recognise and respond to security threats
- ○ **C.** To help employees type faster
- ○ **D.** To train employees to repair computer hardware

Q21 (2 marks)

A system uses three authentication factors: a password, a fingerprint scan, and a security token.
What type of authentication is this?

- ○ **A.** Single-factor authentication
- ○ **B.** Two-factor authentication
- ○ **C.** Multi-factor authentication
- ○ **D.** No authentication

Q22 (2 marks)

An attacker creates fake positive reviews for a malicious app to make it appear trustworthy and encourage downloads.
What social engineering principle does this exploit?

- ○ **A.** Social proof — people trust what others appear to endorse

- ○ **B.** Data compression — reducing file sizes
- ○ **C.** Defragmentation — reorganising storage
- ○ **D.** Load balancing — distributing traffic

Q23 (2 marks)

A company maintains a schedule for testing and deploying security updates across all systems within 30 days of release.
What is this practice called?

- ○ **A.** Patch management
- ○ **B.** Social engineering
- ○ **C.** Typosquatting
- ○ **D.** Credential stuffing

Q24 (3 marks)

A major software vendor's update server is compromised. Attackers insert malware into a legitimate software update, which is then automatically installed on thousands of customer systems worldwide. What type of attack is this, and why is it particularly dangerous?

- ○ **A.** A supply chain attack; it is dangerous because victims trust and automatically install updates from the legitimate vendor, bypassing normal security scrutiny
- ○ **B.** A brute force attack; it is dangerous because it tries many passwords
- ○ **C.** A denial of service attack; it is dangerous because it makes websites unavailable

○ **D.** Shoulder surfing; it is dangerous because attackers watch people type

Q25 (3 marks)

A company implements a security model where:
• No user or device is automatically trusted, even inside the network

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• Every access request is verified regardless of location
• Users receive minimum necessary access for each specific task
```

What security model is this, and why is it increasingly adopted?

○ **A.** Open access model; it is adopted because it makes systems easier to use

○ **B.** Zero trust architecture; it is adopted because traditional perimeter security is insufficient when users work remotely, use cloud services, and threats can originate from inside the network

○ **C.** No security model; it is adopted to reduce costs

○ **D.** Single sign-on; it is adopted to reduce password fatigue

Q26 (3 marks)

An attack executes malicious code entirely in computer memory without writing any files to the hard drive.
Why does this technique make detection more difficult?

○ **A.** Fileless attacks are easier to detect because they use more memory

○ **B.** Traditional antivirus relies heavily on scanning files on disk; fileless malware leaves no files to scan, evades signature detection, and disappears when the system restarts unless it establishes persistence through other means

○ **C.** Fileless attacks only work on mobile phones

○ **D.** Fileless attacks are always visible to users on screen

Q27 (3 marks)

During a security incident, the response team follows these steps:
• Identification: Confirm an incident has occurred

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• Containment: Isolate affected systems to prevent spread
• Eradication: Remove the threat from all systems
• Recovery: Restore systems to normal operation
• Lessons learned: Analyse the incident to improve future respons
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Why is following a structured incident response process important?

○ **A.** Structured processes are slower and less effective than improvised responses

○ **B.** Incident response processes are only needed for minor security issues

○ **C.** Following a process guarantees no security incidents will ever occur

○ **D.** The process ensures thorough handling of incidents — rushed responses may miss threat persistence, contaminate evidence, cause unnecessary damage, or fail to prevent recurrence; each phase has specific objectives that ad-hoc responses often skip

Q28 (3 marks)

A security assessment reveals:
• Technical controls (firewalls, antivirus, encryption) are well-implemented

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• 60% of employees use the same password for work and personal ac
• Password reset requests are processed over the phone without ic
• USB drives found in the car park have been inserted into work c
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What does this assessment reveal about the organisation's security posture?

- ○ **A.** The organisation has excellent security with no vulnerabilities
- ○ **B.** Technical controls are strong but human factors create significant vulnerabilities; password reuse enables credential stuffing, unverified password resets enable social engineering, and USB insertion enables baiting attacks — comprehensive security requires addressing both technical and human elements
- ○ **C.** Only technical security matters; human behaviour is irrelevant
- ○ **D.** The organisation should remove all technical controls

Q29 (3 marks)

A company implements the 3-2-1 backup strategy:
• 3 copies of data (original plus two backups)
• 2 different storage media types (e.g., hard drive and cloud)
• 1 copy stored off-site
How does this strategy protect against different types of data loss?

- ○ **A.** The strategy provides no protection; one backup is sufficient
- ○ **B.** The strategy only protects against accidental deletion
- ○ **C.** Off-site backups are unnecessary if local backups exist
- ○ **D.** Three copies protect against single-point failures; two media types protect against technology-specific failures (e.g., all hard drives failing simultaneously); off-site storage protects against site-wide disasters (fire, flood) and ransomware that encrypts network-connected storage

Q30 (3 marks)

A security analyst warns: 'AI-powered attacks are becoming more sophisticated. Attackers use machine learning to craft more convincing phishing emails, automate vulnerability discovery, and evade detection systems.'
Which assessment of this emerging threat is correct?

- ○ **A.** AI cannot be used for cyber attacks; this concern is unfounded
- ○ **B.** AI-powered attacks represent a genuine evolution in the threat landscape; machine learning enables more personalised phishing, faster vulnerability exploitation, and adaptive malware that learns to evade defences — organisations must adopt AI-enhanced defences and update security strategies
- ○ **C.** Current security measures are already sufficient against any AI-powered attacks
- ○ **D.** Disconnecting from the internet is the only solution to AI threats

Total: **55** marks • Time: **30 min**

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